Python Practice Questions
(Indexing, Slicing, enumerate, zip, Tuple, Set)

Instructions

Solve the following questions. Each question includes a **hint** to guide you.

Indexing & Negative Indexing

1. Given $l = [10, 20, 30, 40, 50]$, print the first element. **Hint:** Indexing starts from 0.
2. Print the last element using negative indexing. **Hint:** Use index -1.
3. What is the output?

   ```
   l = [5, 6, 7, 8]
   print(l[-2])
   ```

   Hint: Count backward from the end.
4. Print the middle element of:

   ```
   nums = [1, 3, 5, 7, 9]
   ```

   Hint: Find the index of the middle value.
5. Given `names = ["ram", "shyam", "hari", "geeta"]`, print `"hari"`. **Hint:** Identify its position.
6. What happens if you try to access an index that does not exist? **Hint:** Think about Python errors.
7. Print the second last element of a list. **Hint:** Use negative indexing.
8. Predict the output:

   ```
   l = [1, 2, 3, 4]
   print(l[-1], l[-4])
   ```

   Hint: Both are valid indices.

Slicing

9. Create a sublist `[1, 2, 3]` from:

```
l = [1, 2, 3, 4, 5]
```

   Hint: Use start and stop indices.

10. Slice the last three elements of the list. **Hint:** Use negative slicing.

11. What is the output?

```
print(l[:2])
```

   Hint: Start index is omitted.

12. What does this return?

```
print(l[1:4])
```

   Hint: Stop index is excluded.

13. Copy a list using slicing. **Hint:** Use full slice.

14. Remove the first and last elements using slicing. **Hint:** Skip boundary indices.

15. Predict the output:

```
l = [1, 2, 3, 4, 5]
print(l[-3:-1])
```

   Hint: Stop index is not included.

16. What is the difference between `l` and `l[:]`? **Hint:** Think about reference vs copy.

enumerate()

17. Print each item with its index using `enumerate()`. **Hint:** Loop over enumerate.

18. Start enumeration from 1. **Hint:** Use the `start` parameter.

19. Predict the output:

```
users = ["ram", "hari"]
print(list(enumerate(users)))
```

   Hint: enumerate returns tuples.

20. Print output in the format:

```
1 ram
2 hari
```

Hint: Use unpacking.

21. Why is `enumerate()` better than `range(len())`? **Hint:** Readability and safety.

22. Print users at even positions using enumerate. **Hint:** Check index condition.

zip()

23. Zip two lists and print paired values. **Hint:** Use a for-loop.

24. Predict the output:

```
names = ["ram", "shyam"]
phones = ["98", "97", "96"]
print(list(zip(names, phones)))
```

Hint: zip stops early.

25. Print output as:

```
ram -> 9812
shyam -> 9813
```

Hint: Unpack zip values.

26. What happens when list lengths are different? **Hint:** Observe the shortest list.

27. Convert zipped data into a list. **Hint:** Use `list()`.

Tuple

28. Create a tuple and print its first element. **Hint:** Tuples support indexing.

29. What error occurs here and why?

```
t = (1, 2, 3)
t[1] = 5
```

Hint: Think immutability.

30. Convert a list into a tuple. **Hint:** Use constructor.

31. Unpack the tuple:

```
data = (1, "ram")
```

Hint: Match variable count.

32. Use enumerate with a tuple. **Hint:** Tuples are iterable.

Set

33. Create a set from a list with duplicate values. **Hint:** Sets remove duplicates.
34. Why does a set not maintain order? **Hint:** Think hashing.
35. Add one element to a set. **Hint:** Use `add()`.
36. Add multiple elements using `update()`. **Hint:** Pass an iterable.
37. Remove an element from a set. **Hint:** Use `remove()`.
38. Find common elements between two sets. **Hint:** Intersection operation.
39. Find elements only in the first set. **Hint:** Difference operation.
40. Write a program to store hobbies without duplicates. **Hint:** Use a set and loop.